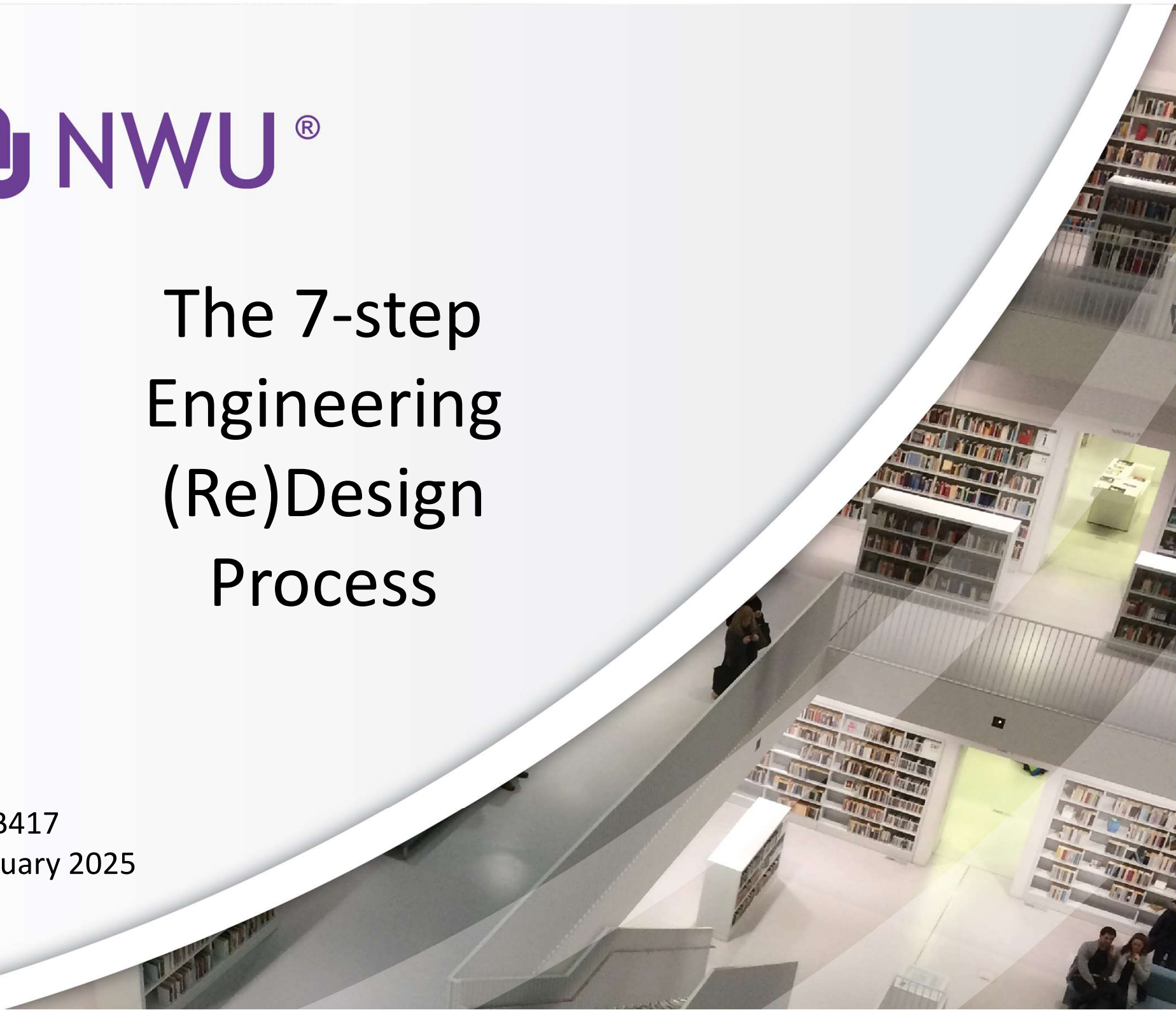




The 7-step Engineering (Re)Design Process

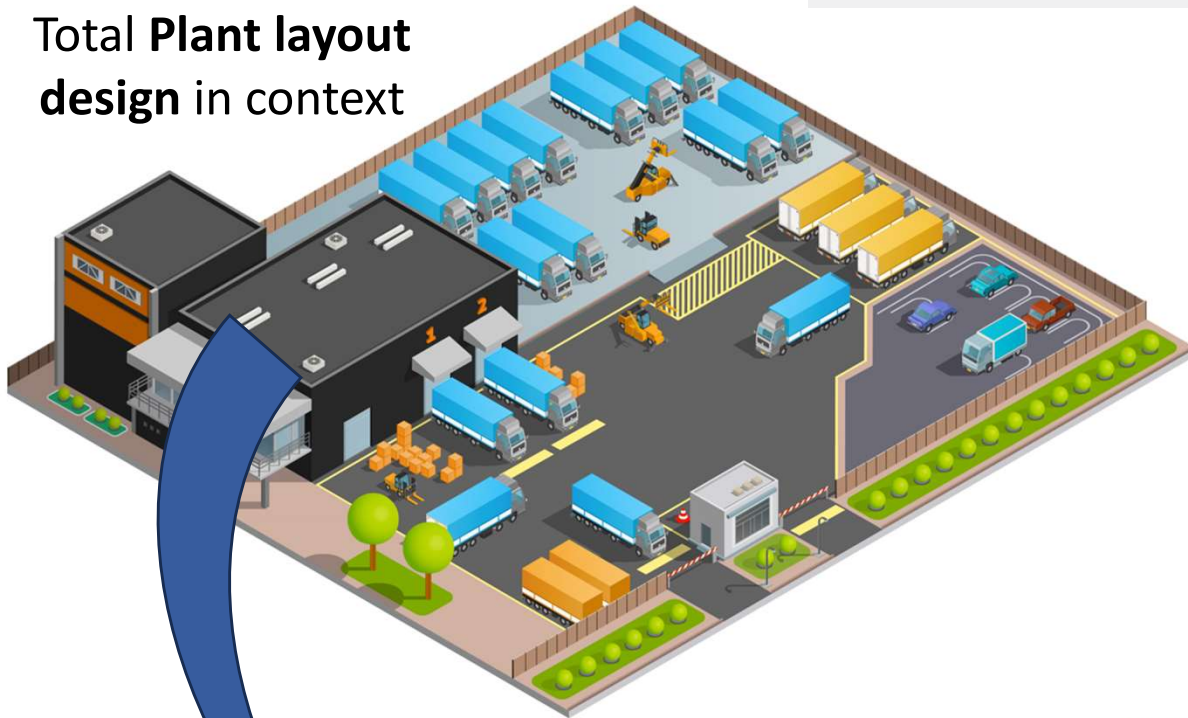
INGB417
February 2025



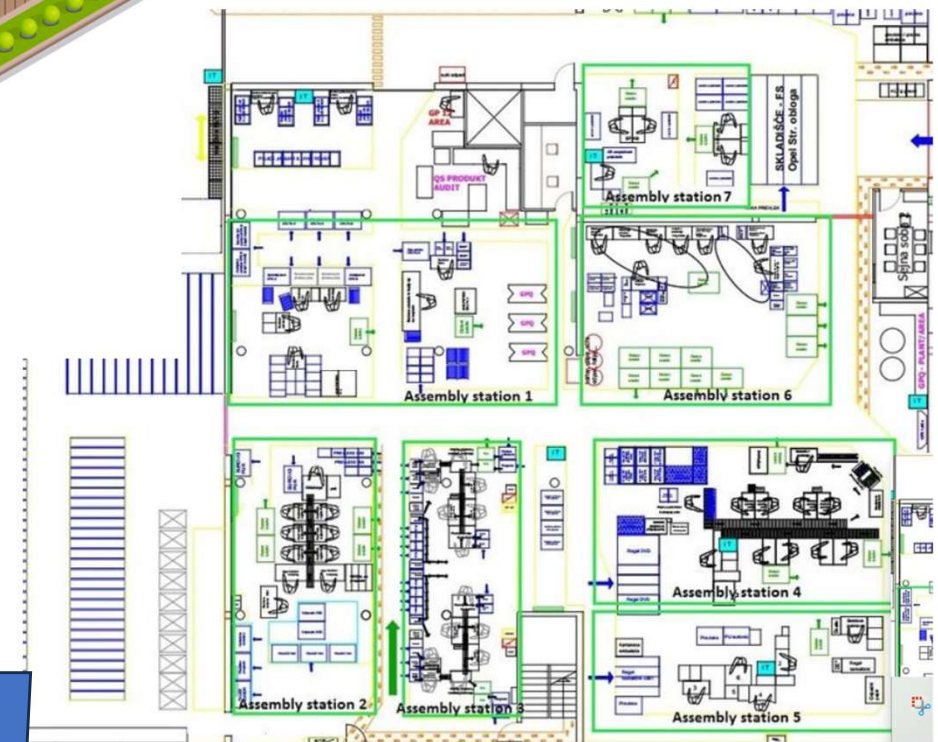
Facilities Design

always include 2 'designs'

Total Plant layout design in context



Detail production process layout = **Manufacturing Process design**

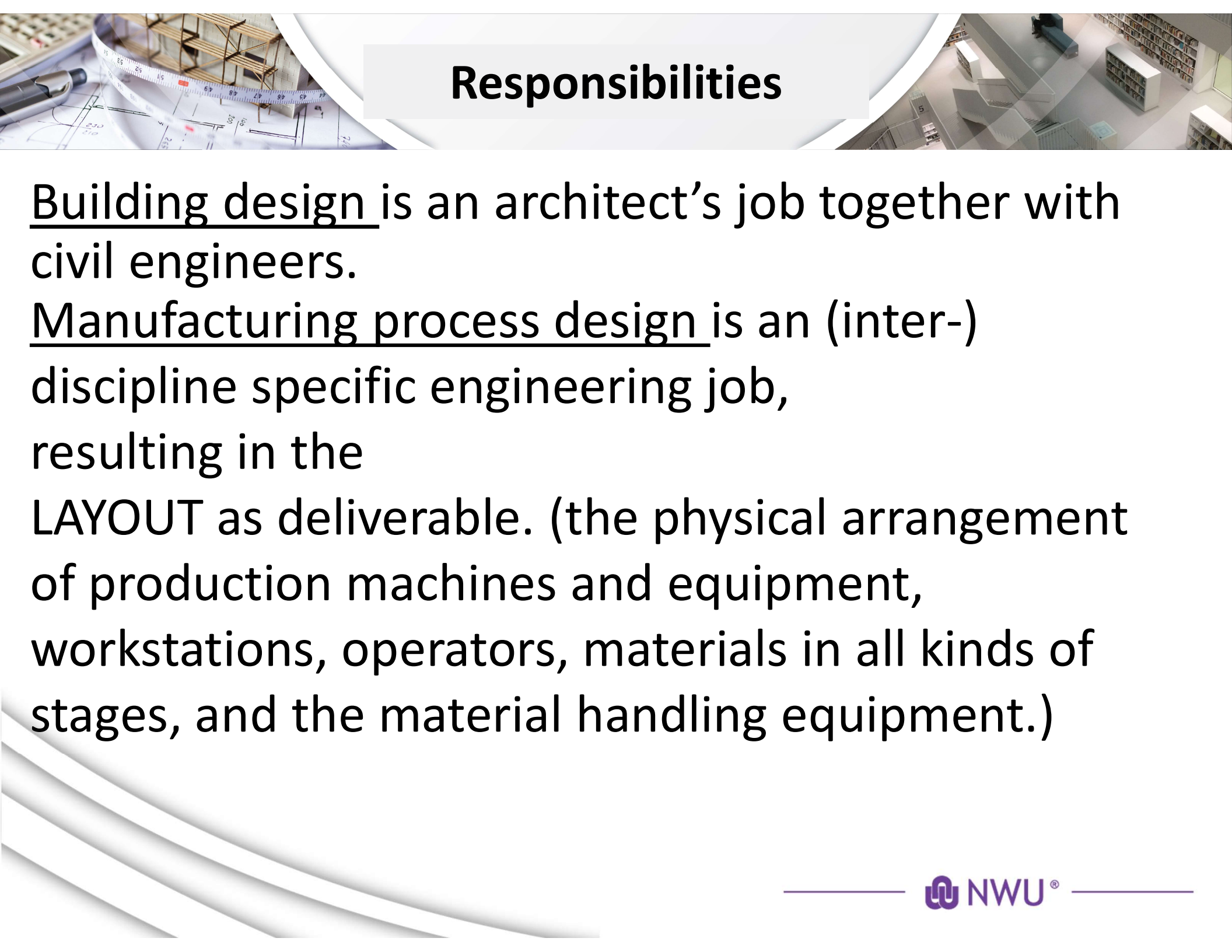




Definitions

Facilities design is the *organization* of a company's physical assets (buildings, equipment, materials) to promote efficient use of resources. Facilities design therefore includes location, plant/factory *layouts* and *material handlings systems*.

Manufacturing process design includes designing the products, then designing the required *manufacturing process activities* to make these products. (this may involve several options and many interrelationships between activities)



Responsibilities

Building design is an architect's job together with civil engineers.

Manufacturing process design is an (inter-) discipline specific engineering job, resulting in the LAYOUT as deliverable. (the physical arrangement of production machines and equipment, workstations, operators, materials in all kinds of stages, and the material handling equipment.)

The 7 steps : Engineering Design process

Step 1: DEFINE the Design Problem / improvement opportunity



Step 2: ANALYZE the Problem



Step 3: Determine Design REQUIREMENTS / Specifications



Step 4: Develop ALTERNATIVE Designs



Step 5: EVALUATE the Alternatives



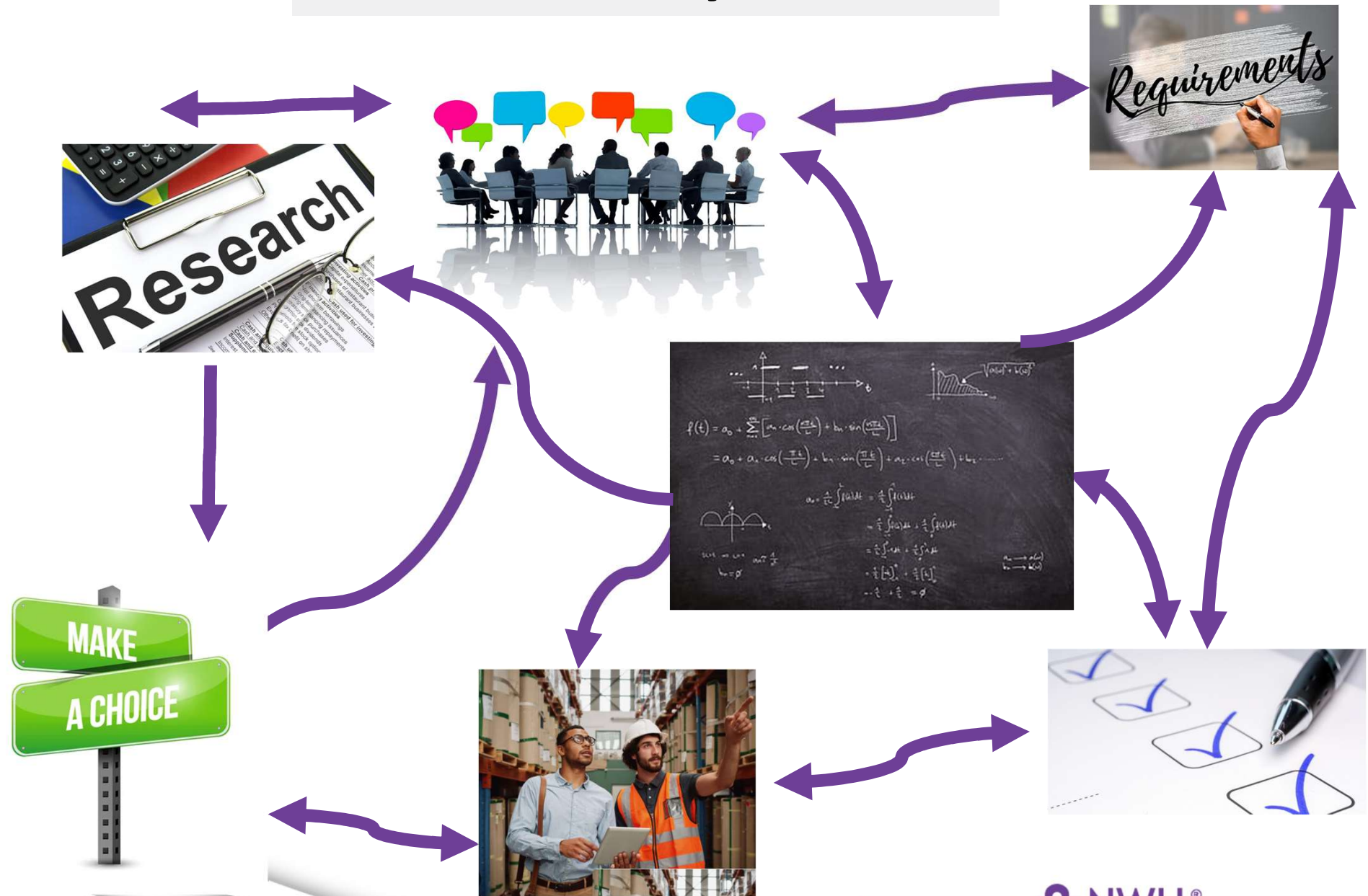
Step 6: CHOOSE Solution(s)

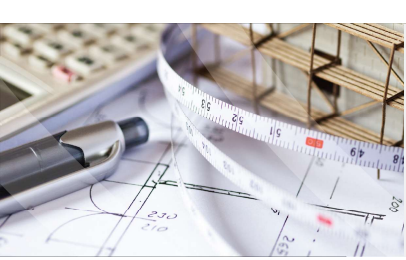


Step 7: PRESENT the Design

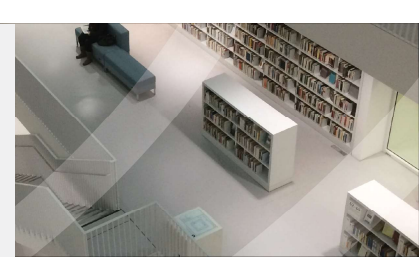
The neat
structured
theoretical
way

The Design Process reality

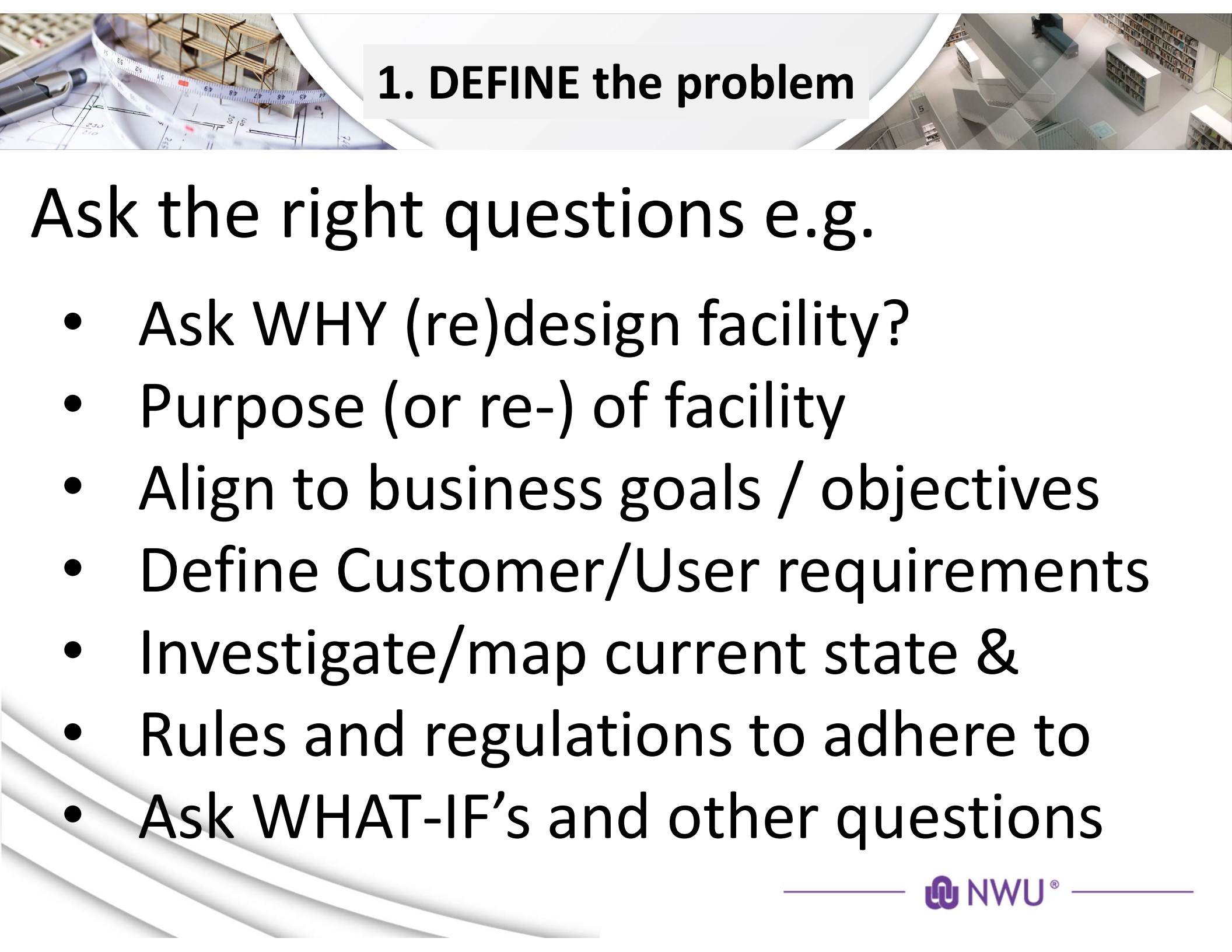




1. DEFINE the (design) “problem”



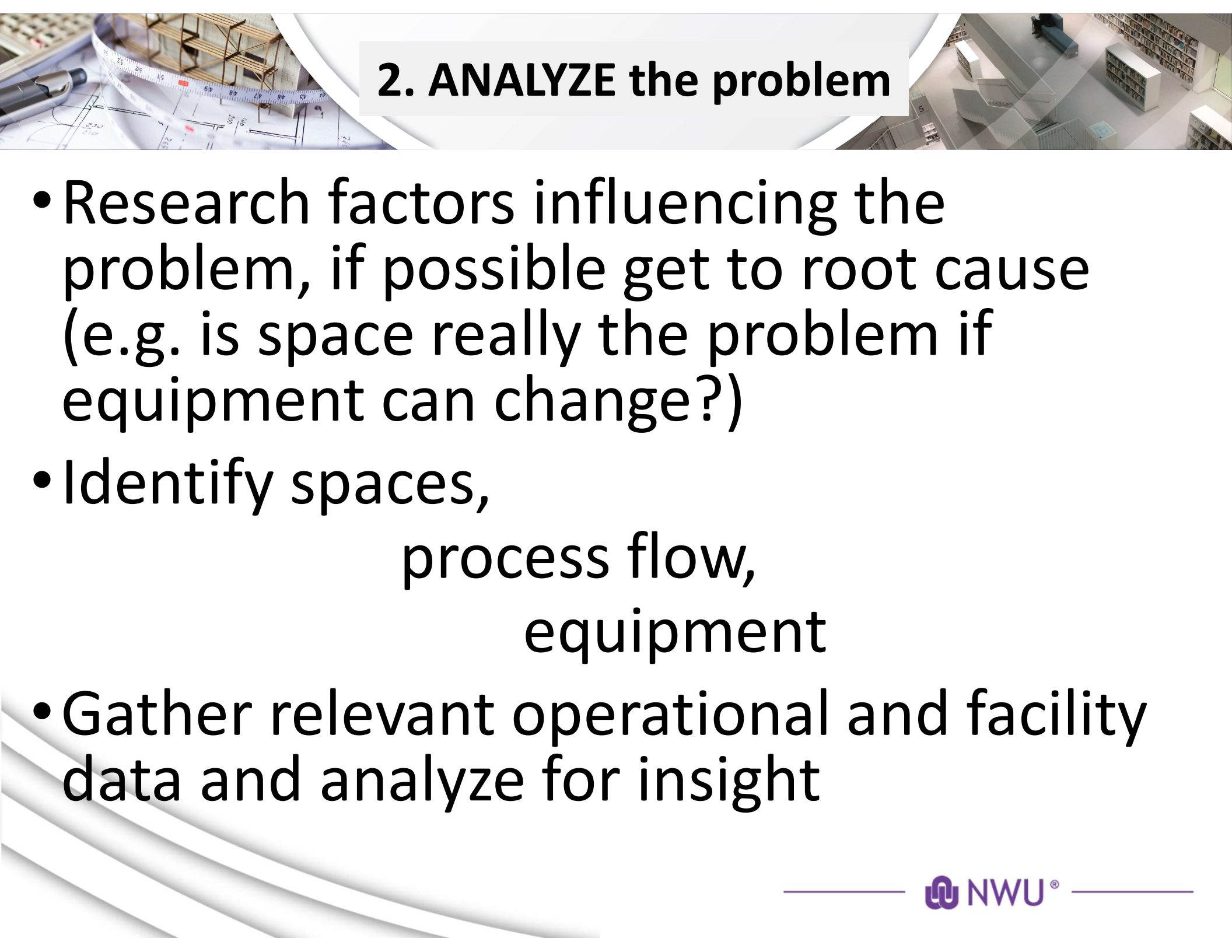
- Identify issues (e.g. inefficiencies, frustrations, or improvement opportunities and needs)
- Understand the goals, the scope, assumptions, constraints.



1. DEFINE the problem

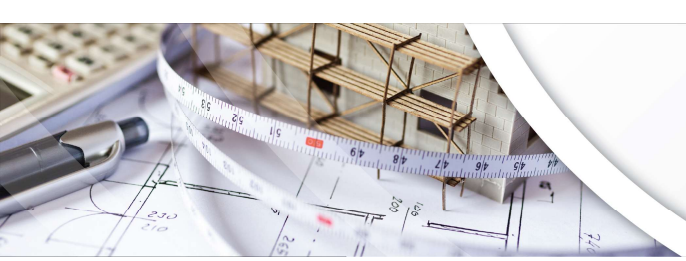
Ask the right questions e.g.

- Ask WHY (re)design facility?
- Purpose (or re-) of facility
- Align to business goals / objectives
- Define Customer/User requirements
- Investigate/map current state &
- Rules and regulations to adhere to
- Ask WHAT-IF's and other questions

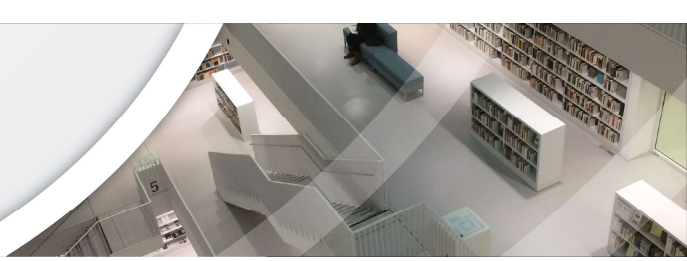


2. ANALYZE the problem

- Research factors influencing the problem, if possible get to root cause (e.g. is space really the problem if equipment can change?)
- Identify spaces,
process flow,
equipment
- Gather relevant operational and facility data and analyze for insight

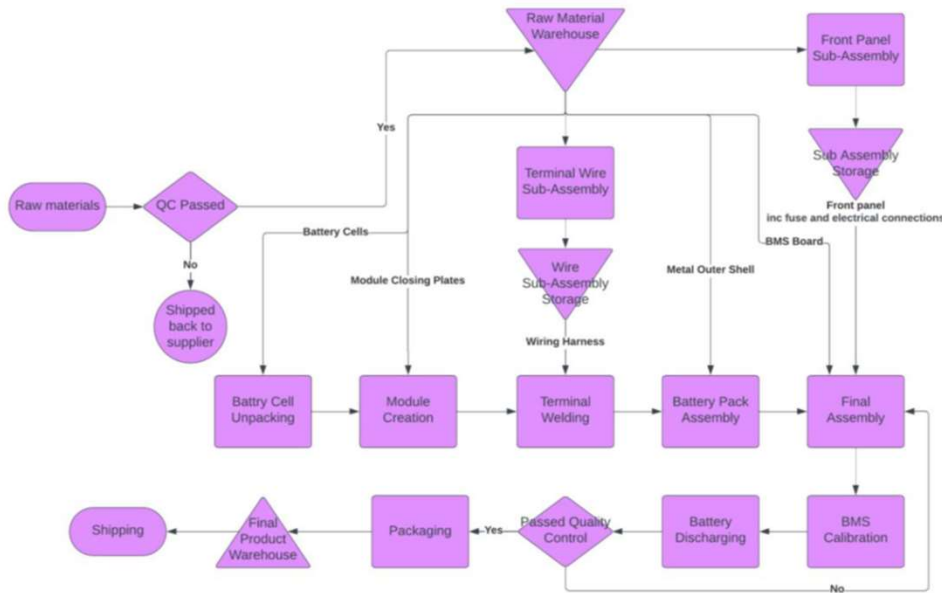


Use IE Tools and techniques



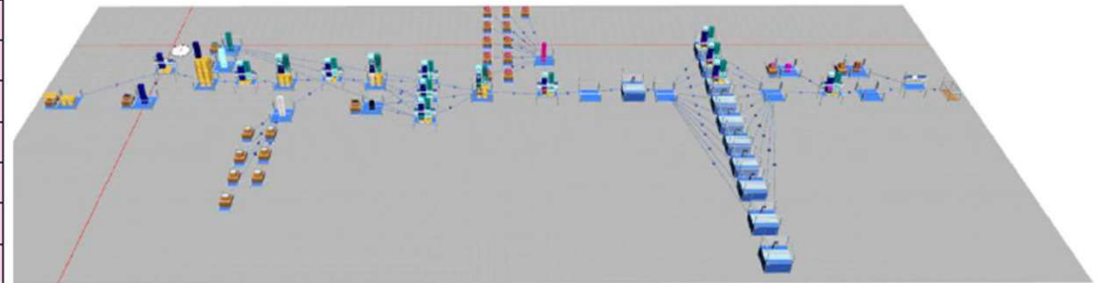
- Physical *observations*
- Talking to people (operators/managers) [sometimes this is called a Gemba walk, but it is just a *plant tour*]
- Current state measurements (*layout and station dimensions*)
- *Time studies* as necessary
- *Process flows* mapping (or ValueStreamMapping)
- Get detail on process times, to determine cycle times,
- Develop *simulation* models if it will help
- Do Capacity calculations
- Get machine specifications, dimensions

Use IE Tools and techniques



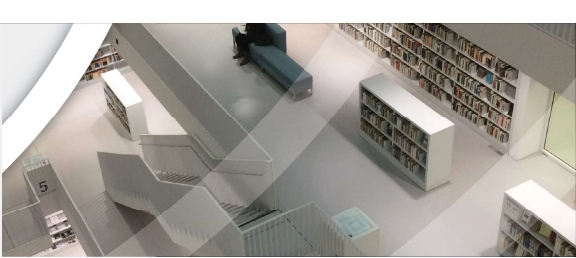
Current Spacing Per Department	
Departments	Spacing Required(m^2)
Raw Materials Warehouse	219
Mechanical	532
Front Panel Sub-Assembly	185
Wiring Department	55
Battery Assembly	402
Final Product Warehouse	247.6
Auxiliary	731
Total	2371.6

Travel Time Between Departments		
From	To	Duration
Raw Material Warehouse	Mechanical Department	01:40
Raw Material Warehouse	Front Panel Sub-Assembly	02:11
Raw Material Warehouse	Start of Battery Assembly	02:57
Front Panel Sub-Assembly	Start of Battery Assembly	00:50
Front Panel Sub-Assembly	Mechanical Department	00:35
Mechanical	Start of Battery Assembly	01:08
From	To	Duration
Start of Battery Assembly	Repair	00:29
End of Battery Assembly	Final Product Warehouse	00:21
Final Product Warehouse	Long Term Warehouse	03:07



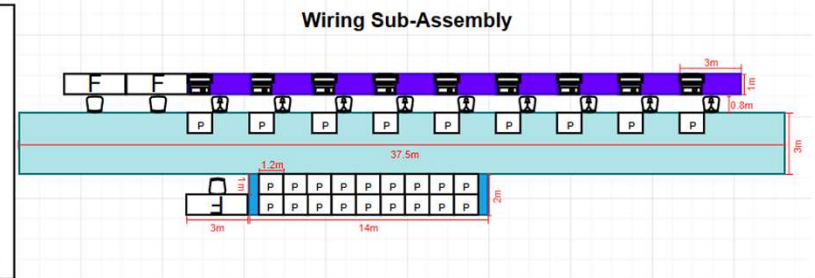
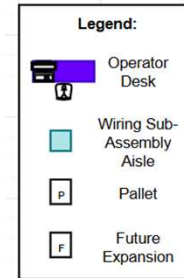
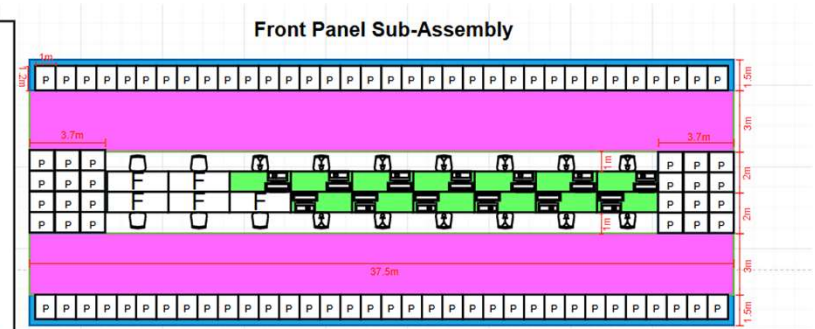
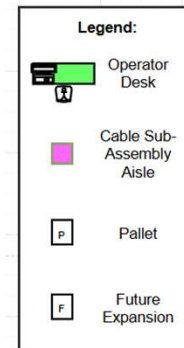
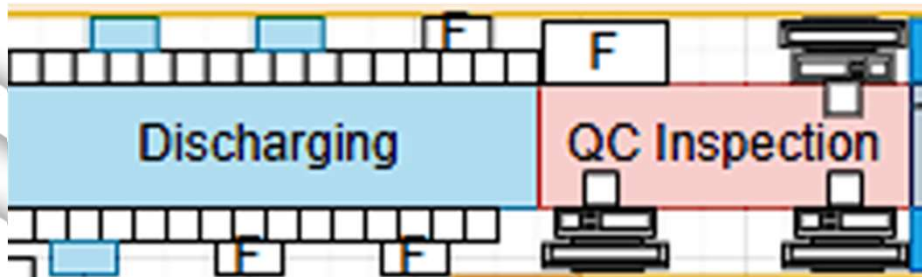
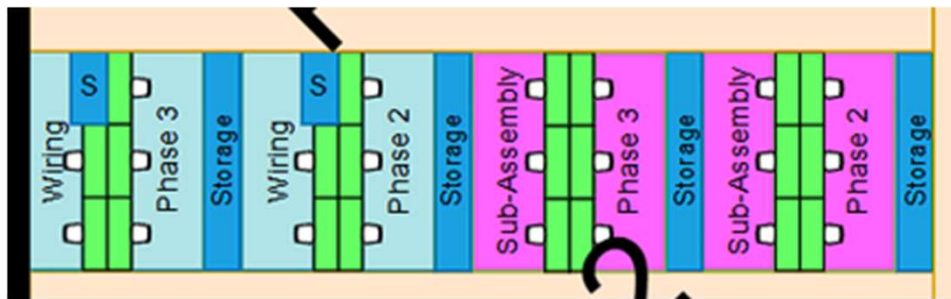
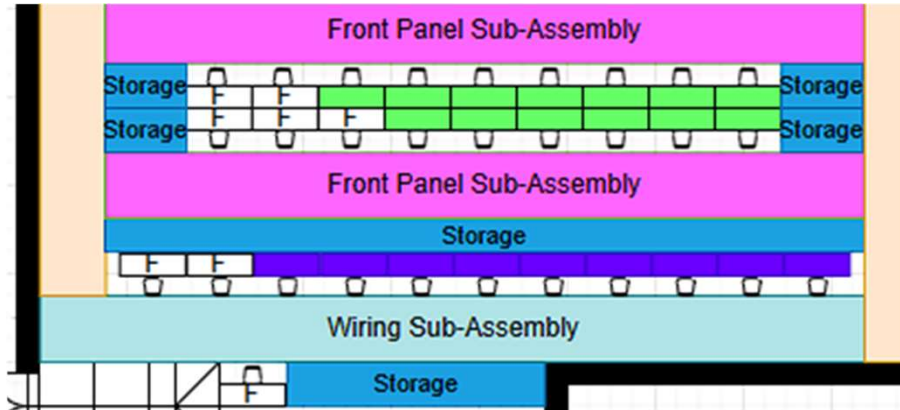


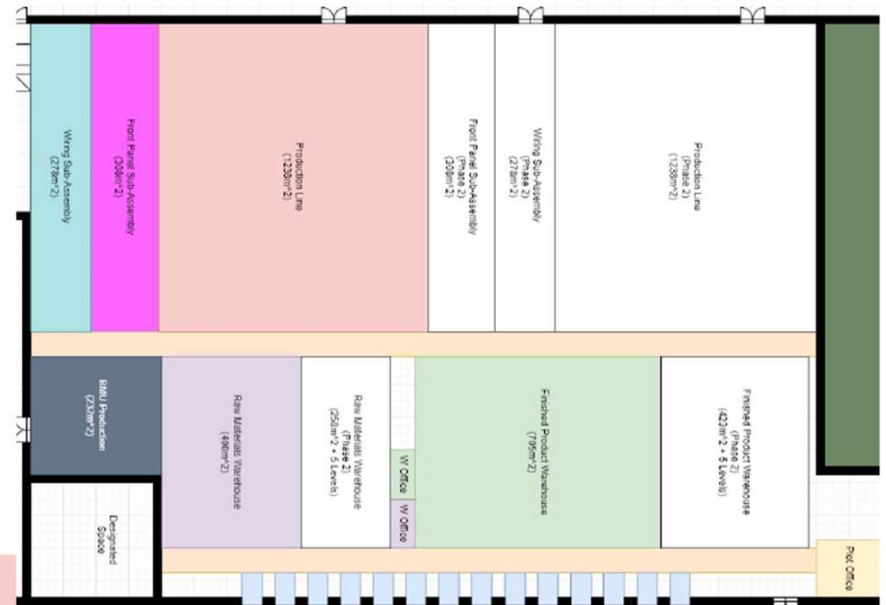
3. Determine DESIGN REQUIREMENTS



- Focus on space and functional needs for all the relevant things happening in a facility eg (for production factory):
 - Raw materials / parts receiving, storage
 - Production process activities, workstations, departments, operators, equipment, material handling strategies, Work-in-progress storage
 - Finished goods storage, dispatch
 - All administrative and management personnel offices
 - Parking for staff, deliveries and customers
 - Security

3. Do Detail DESIGN from ground up





-
- Production Line (138m²)
- Front Panel Sub-Assembly (250m²)
- Back Panel Sub-Assembly (250m²)
- Raw Materials Warehouse (400m²)
- Finished Product Warehouse (700m²)
- W Office
- Port Office

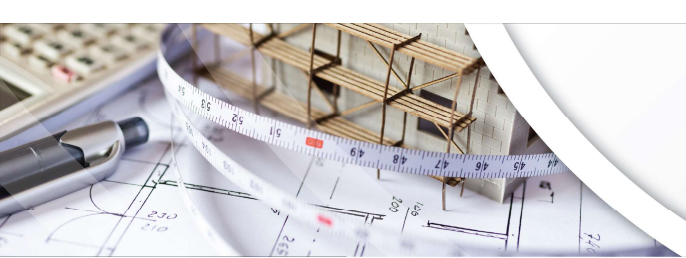
5. EVALUATE the Alternatives

EVALUATION	WEIGHT	Factory Layout					
		Alt. 1		Alt. 2		Alt. 3	
		Rating	Score	Rating	Score	Rating	Score
Theoretical Efficiency	0.2	6	1.2	5	1	4	0.8
Implementation Cost	0.3	6	1.8	5	1.5	6	1.8
Space Utilisation and optimisation	0.3	3	0.9	6	1.8	7	2.1
Scalability	0.1	2	0.2	5	0.5	7	0.7
Ergonomics	0.1	6	0.6	5	0.5	6	0.6
Total	10	4.7		5.3		6	

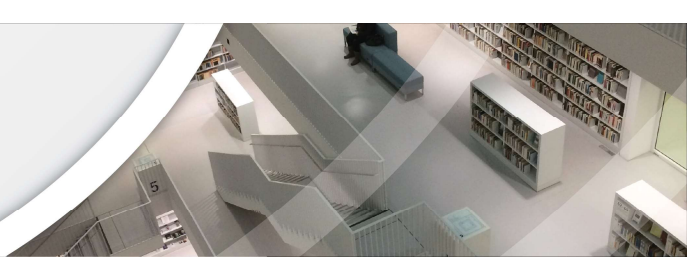
- Compare solutions based on feasibility, cost, effectiveness or other criteria
- Also consider sustainability
- Ideally get input and feedback from customer

Table 22: Pros and Cons Analysis

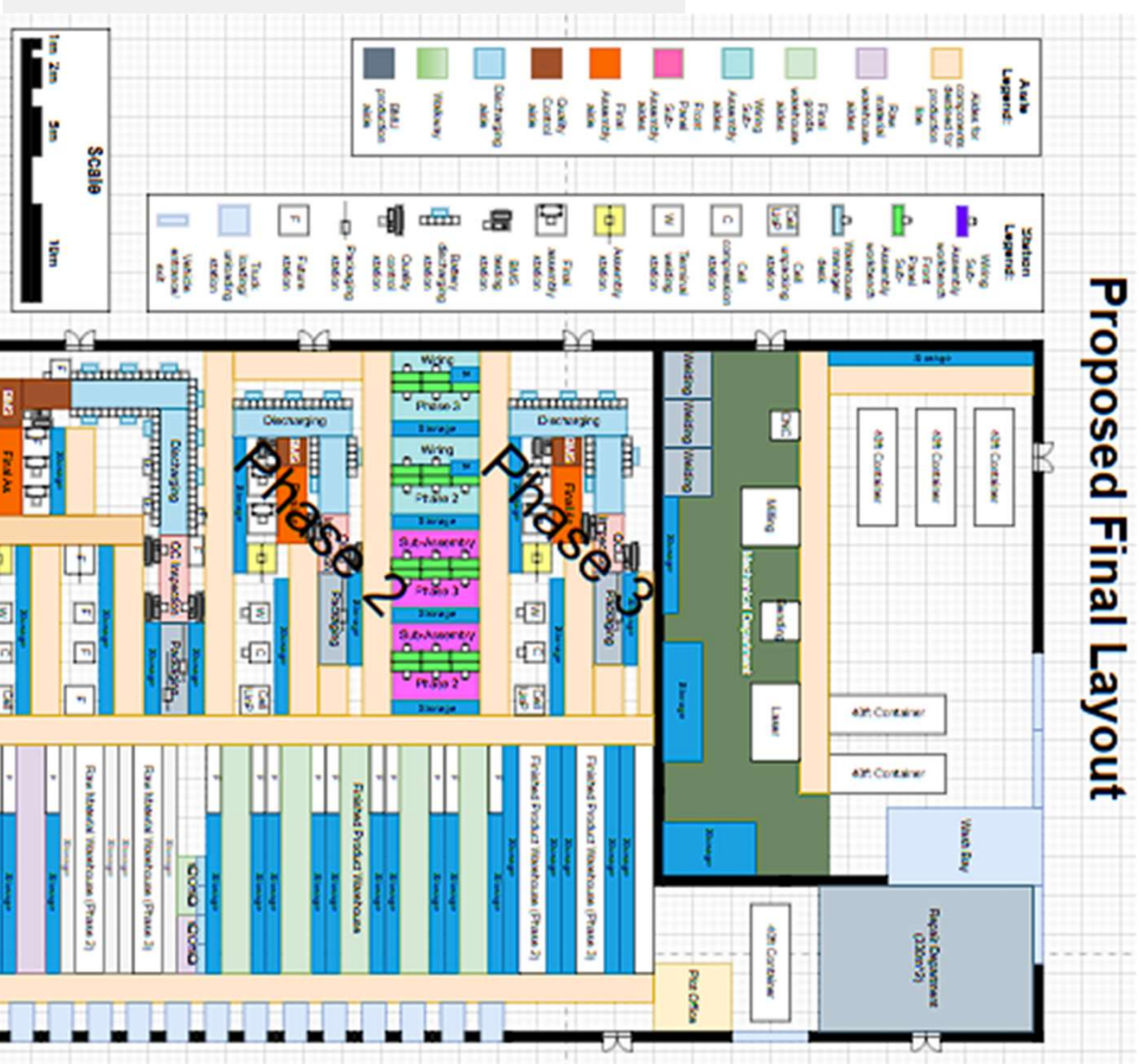
Alternative Type	Pros	Cons
Alternative 1	This layout was optimised for one goal: to produce 7200 units per month in the most efficient way possible.	No plans for expansion above the 30% over 5 years that will lead to a factory shut-down when expansion is necessary.
Alternative 2	Expansion plans were implemented in this alternative	Due to the bigger material handling distances, the layout is



6. CHOOSE (a) design(s)

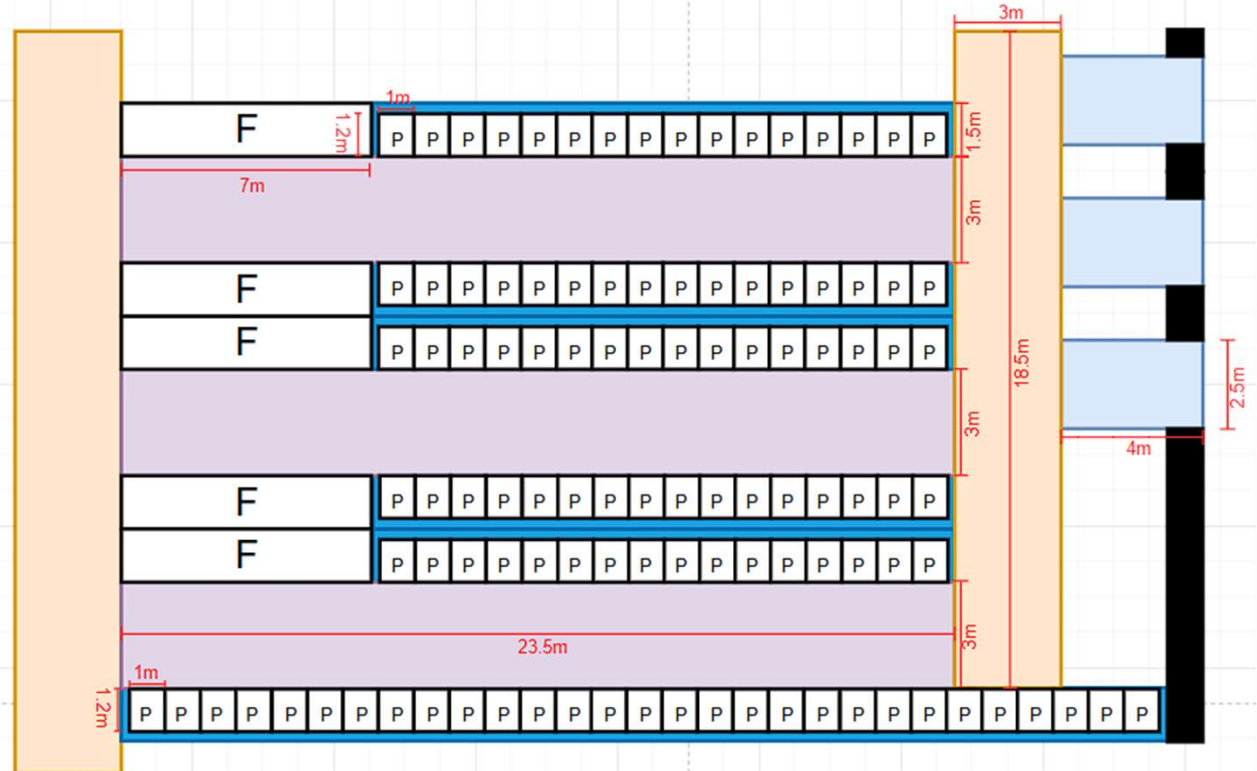
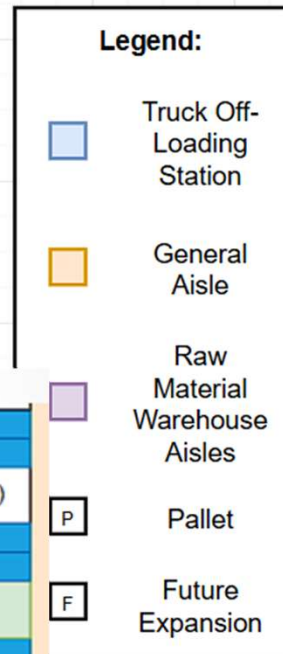


- Ensure the solution meet the design requirements
- Involve customer to ensure alignment of expectations
- Justify the choice with DATA and reasoning



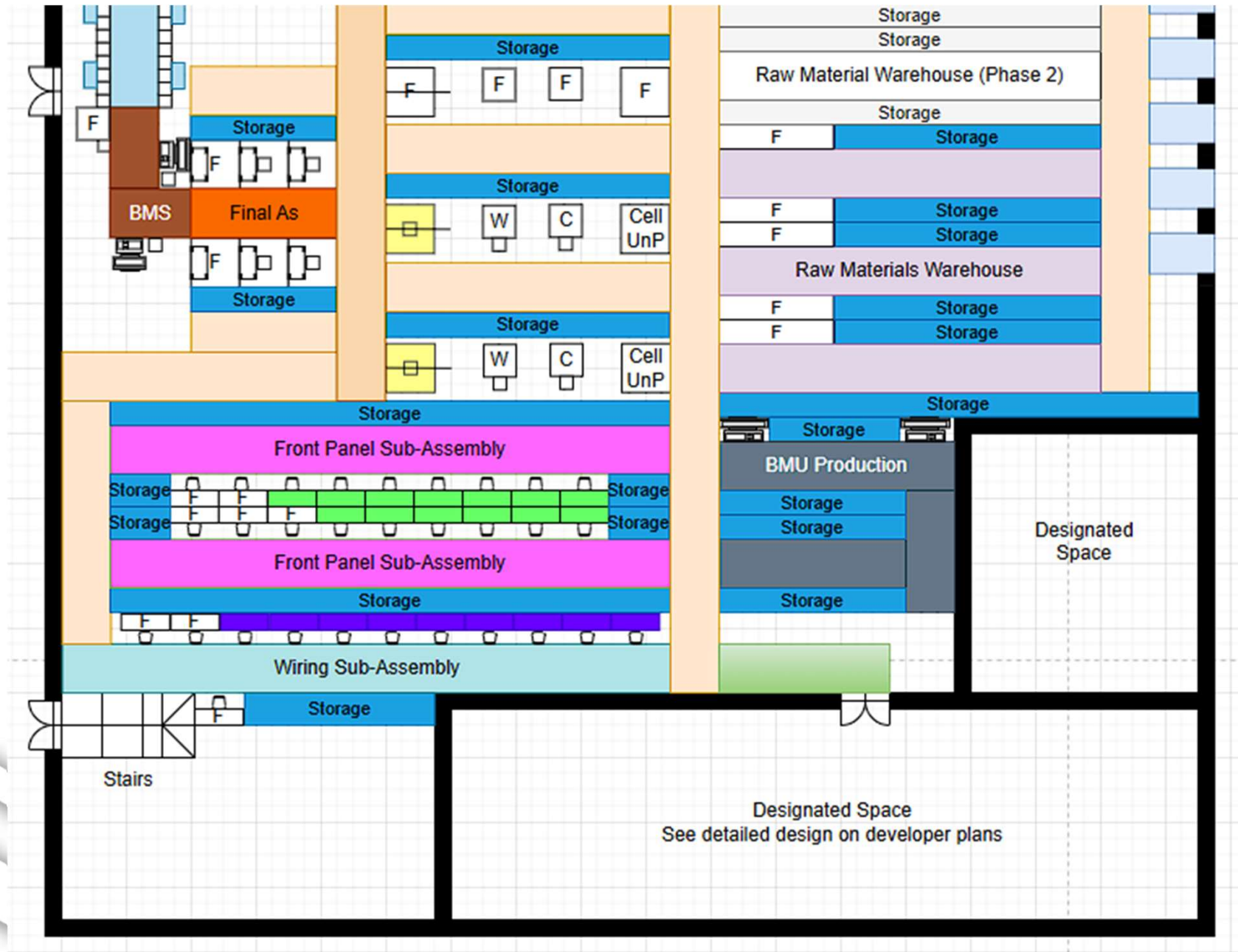
Detailed Design

Raw Material Warehouse



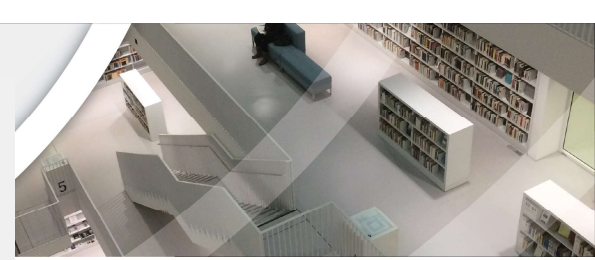
Note dimensions strategically shown

Master Facility Plan





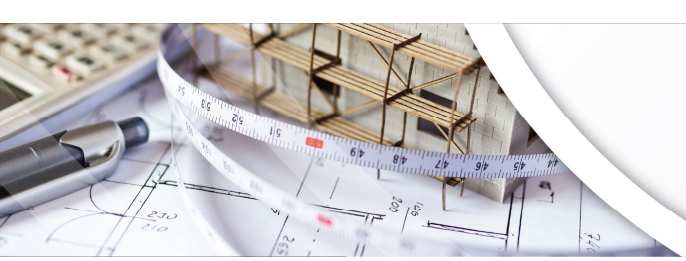
7. PRESENT the design ie. show the MFP



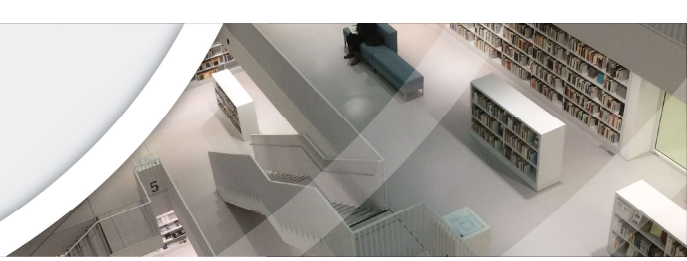
Present the plans
informally to
stakeholders during
process, but then also

Present the Master
Facility Plan & Business
Case formally for
management approval





Design Acceptance = formal milestone



Once a design has been FORMALLY ACCEPTED = Signed off by the relevant authority, then the project is entering **Build, Commission and Handover** phase

BUYING, BUILDING and SETUP can start = completely different project in a way

When that is finished there should be a FORMAL HANDOVER to OPERATIONS to start work.

